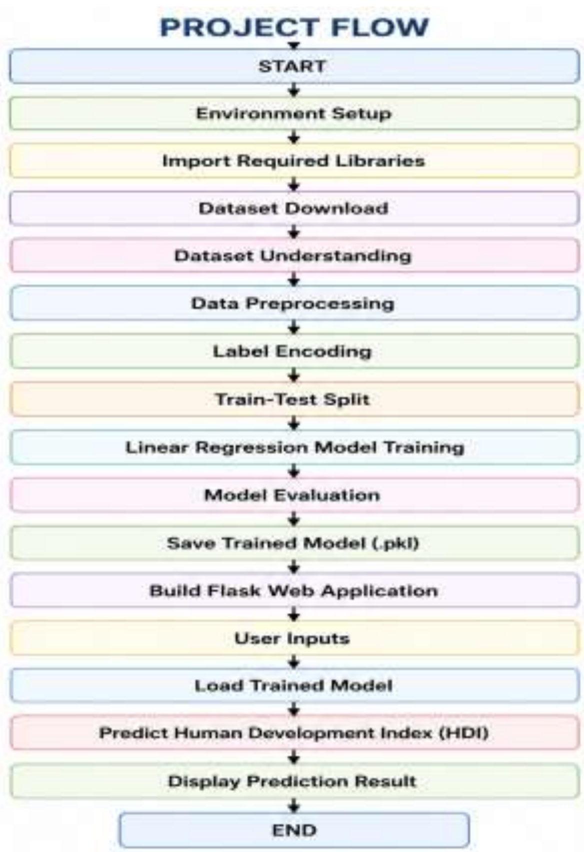
Project Workflow

Date	27 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being

The project is developed through a structured workflow consisting of multiple epics, covering data collection, analysis, preprocessing, model development, and deployment. Each epic focuses on a specific stage of the machine learning lifecycle to ensure systematic and efficient project execution.



Epic 1: Environment Setup and Package Installation

Story 1: Download and install all required Python packages, machine learning libraries, and Flask dependencies needed for the project.
Story 2: Create and organize the project folder structure, including directories for datasets, models, templates, static files, and source code.

Epic 2: Importing Required Libraries

Story 1: Import all necessary libraries and modules such as NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Pickle, and Flask to support data analysis, model development, and web application creation.

Epic 3: Dataset Download and Understanding
Story 1: Download the dataset from Kaggle and load it into the development environment.
Story 2: Explore and understand the dataset structure, features, target variable, and data types.
Story 3: Perform data visualization to identify patterns, trends, distributions, and relationships within the dataset.

Epic 4: Data Preprocessing and Label Encoding

Story 1: Select dependent and independent variables required for model training.
Story 2: Check for missing values and handle null entries appropriately to improve data quality.
Story 3: Perform label encoding to convert categorical text values into numerical representations suitable for machine learning algorithms.
Story 4: Prepare the cleaned and transformed dataset for model development.

Epic 5: Dividing the Dataset into Train and Test Data

Story 1: Split the processed dataset into training and testing sets to enable model learning and performance evaluation on unseen data.

Epic 6: Fitting the Model

Story 1: Train the Linear Regression model using the prepared training dataset.
Story 2: Generate predictions using the trained model and analyze the prediction results.
Story 3: Evaluate model performance using appropriate regression metrics and visualizations.

Epic 7: Saving the Model

Story 1: Save the trained Linear Regression model using serialization techniques such as Pickle so it can be reused without retraining.

Story 2: Store the model file for deployment and future prediction purposes.

Epic 8: Building the Flask Web Application

Story 1: Develop the Flask backend to handle user requests, process inputs, load the trained model, and generate predictions.
Story 2: Create HTML templates and integrate them with Flask to provide a user-friendly interface.
Story 3: Run, test, and validate the complete web application to ensure accurate predictions and smooth functionality.